

You are implementing the backend for the Redrob AI Candidate Ranking System.

The uploaded Backend API document is the primary functional specification.

Read the entire document carefully before making any changes.

IMPORTANT:

The document defines the intended features and workflows.

However, the following implementation rules take precedence wherever they differ from the document.

CORE PRINCIPLE

DO NOT replace the existing local AI pipeline.

Instead, extend the project to support two execution modes.

The application must support BOTH:

1. Competition Mode
2. API Mode

Both modes must coexist permanently.

Competition Mode remains the default.

COMPETITION MODE

Competition Mode must remain exactly as required by the hackathon.

Characteristics:

- No external APIs
- No internet dependency
- Fully deterministic
- CPU-only execution
- Reproducible results

Pipeline:

Load Candidates

↓

Title Filter



Bi-Encoder Semantic Search



Feature Extraction



Skill Scoring

Career Analysis

Experience

Location

Education

Behavioral Signals



Composite Score



Dynamic Shortlist



Cross-Encoder Re-ranking



Reasoning Generator (Template Based)



submission.csv

Competition Mode must never call Gemini, OpenAI, Claude, Cohere, or any external AI service.

API MODE

API Mode is an enhancement layer.

DO NOT replace the local ranking pipeline.

The local ranking pipeline remains authoritative.

API Mode may enhance only the following:

- ✓ Job Description Parsing
- ✓ Candidate Explanation
- ✓ Candidate Comparison
- ✓ Executive Summary
- ✓ Recruiter Chat
- ✓ Interview Question Generation
- ✓ Skill Gap Analysis
- ✓ Hiring Recommendation

Ranking itself must continue using:

- Local Embeddings
- Local Feature Scoring
- Local Cross-Encoder

Do NOT replace these with LLM ranking.

JOB DESCRIPTION PARSING

Competition Mode

Rule-based parser.

API Mode

LLM parser.

Convert any pasted JD into structured JSON.

REASONING

Competition Mode

Deterministic template reasoning.

API Mode

LLM-generated recruiter-friendly explanation.

CHAT

Implement recruiter chat only in API Mode.

Examples:

Why Candidate #1?

Compare Candidate 3 and 8

Who has Kubernetes experience?

Show candidates with strong leadership

Summarize top 10 candidates

EXECUTIVE SUMMARY

API Mode should generate:

- Hiring Summary
- Candidate Insights
- Risks
- Strengths
- Recommended Interviews

Competition Mode does not require these.

SUPPORTED PROVIDERS

Support only:

- Gemini
- OpenAI

Design the provider layer so more providers can be added later without modifying business logic.

API SETTINGS

Create an API Settings page.

Allow users to:

- Select Provider
- Enter API Key
- Validate Connection
- Enable or Disable API Mode

Store keys securely.

Never hardcode secrets.

FALLBACK

If:

- API key missing
- Rate limit reached
- Network unavailable
- API error

Automatically fall back to Competition Mode.

The application must never fail because of an external API.

PERFORMANCE

Continue using:

- Local sentence-transformer embeddings
- Local feature extraction
- Local cross-encoder reranking
- Candidate embedding cache
- JD embedding cache

Do not replace them with API embeddings.

ARCHITECTURE

Maintain clean separation:

core/

contains

- ranking
- embeddings
- scoring
- reranking

api/

contains

- Gemini
- OpenAI
- Prompt Builders

services/

contains

CompetitionService

ApiEnhancedService

Both services reuse the same ranking engine.

OBJECTIVE

Competition Mode must generate exactly the same submission.csv regardless of API availability.

API Mode should provide a richer recruiter experience without changing the core ranking algorithm.

The ranking engine must have a single source of truth shared by both modes.

Never duplicate ranking logic.

Never replace deterministic ranking with LLM-based ranking.

Only extend the application with optional AI-powered recruiter features.

After implementation:

1. Perform a complete architectural review.
2. Verify that both modes work correctly.
3. Verify Competition Mode produces deterministic outputs.
4. Verify API Mode enhances explanations without affecting rankings.
5. Report any remaining inconsistencies before considering the implementation complete.